

Algebra II with Trigonometry

Chapter 6.4

Olympiad Problems

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Problems

1. Let $f(x) = \tan(2x) + \tan(2x + \pi) + \tan(2x + 2\pi)$. Find the value of $f(\frac{\pi}{8})$.
2. Identify all real x in $[0, 2\pi)$ such that $\cot x = \sec x$.
3. Let S denote the sum of all distinct x in $(0, 2\pi)$ where $\csc x = 2$. Find S .
4. Compute the product of all solutions x in $(0, \pi)$ to the equation $\tan 3x = 0$.
5. Given $y = 3 \tan(4x - \pi)$, determine the distance between two consecutive vertical asymptotes.
6. For $y = \csc kx$ where $k > 0$, suppose that the graph between $x = 0$ and $x = \pi$ displays exactly three vertical asymptotes. Find the value of k .